

Mohan Sai Jyasta

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EDUCATION:

Master's in Electrical Engineering

Aug 2022 – May 2024

San Jose State University, San Jose, CA

GPA: 3.5/4.0

Coursework: Advanced Digital System Design and Synthesis, Advanced Computer Architecture, ASIC CMOS Design, Digital Design with A/DSP, Logic Verification using UVM, SoC Design and Verification with System Verilog.

Bachelor of Technology, Electronics and Communication Engineering

Aug-2017 - July 2021

BML Munjal University, Haryana, India

Coursework: Digital Logic Design (HDL), Electronics-I&II, Computer Organization and Architecture, VLSI Design, Digital Hardware.

SKILLS:

Languages: Verilog, System Verilog, C, Python, Tcl, Shell Scripting (basic)

Concepts & Skills: Design Verification, UVM, SystemVerilog Assertions (SVA), Constrained Random Verification, Coverage (Functional & Code), Formal Verification, RTL Design, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Power-Aware Verification (UPF), SoC and Subsystem Verification, Performance and Power Verification, Post-Silicon Debug, CPU, GPU, DDR Protocol Verification, Cache Verification, Memory Verification IPs (VIPs), RTL-to-GDSII Flow

Tools: Xilinx ISE, Vivado, ModelSim, QuestaSim, Synopsys VCS, IC Compiler, PrimeTime, Cadence Virtuoso, Genus.

Operating Systems: Windows, Linux.

Protocols: MSI, MESI, MESIF, UART, SPI, I2C, AMBA APB, AHB, AXI, PCI.

Soft Skills: Problem Solving, Critical Thinking, Team Player, Time Management, Planning, Quick Learning

INTERNSHIP EXPERIENCE :

Digital Design Intern, TessolveDts, San Jose

July 2024 – Jan 2025

- Developed scalable UVM scoreboards and monitors for verifying Die-to-Die digital interfaces, ensuring accurate data comparison across SerDes link macros and lanes.
- Created reusable UVM-based scoreboards to validate data between the System-on-Chip (SoC) and digital interfaces, enhancing verification efficiency.
- Automated synthesis file preparation using custom Makefiles and Python scripts, improving workflow efficiency and supporting PD drop testing scenarios.
- Wrote System Verilog assertions to verify data integrity across digital interfaces, ensuring correct DUT behavior.

Digital Design Intern (RTL design/verification) | Truechip | India

Dec 2020 - Aug 2021

- Addressed AXI protocol performance verification between multiple master-slave pairs, developing **custom UVM components** and **coverage-driven methodologies** for industrial-grade reliability
 - Employed **constraint-random testing** to generate extensive stimuli variations across protocol states, achieving 25% improvement in AXI protocol verification
 - Enhanced **functional and cross-coverage** by optimizing random stimulus generation using **UVM reusable sequences**, confirming performance stability under diverse operational conditions
 - Automated data processing using **Python** to analyze functional coverage, supporting accelerated reporting cycles
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ACADEMIC PROJECTS:

- RTL Design and Verification of 11-bit Floating Point Adder and Multiplier for AI Engine | VCS**
Implemented an 11-bit floating-point adder and multiplier in System Verilog with 1GHz timing, integrating an AHB bus interface for efficient data transfer. Designed a UVM testbench to verify rounding precision and denormalized numbers and leveraged Python and machine learning for test case coverage prediction.
 - DDR Read/Write Integrity Verification| Synopsys VCS**
Designed and implemented a UVM-based testbench for DDR memory controller verification using Synopsys VCS, simulating sequential and burst read/write transactions while ensuring data integrity and timing compliance through comprehensive checks on written and read data patterns
 - FFT Design and Verification for OFDM | System Verilog, Synopsys VCS, DC Compiler, Python**
Developed a synthesizable 128-point FFT model with a slicing block to decode energy levels in frequency bins. Verified FFT blocking functionality using UVM, integrated with Encoding and IFFT blocks. Created a reference model and calculated coverage metrics to enhance accuracy and testing.
 - Synchronous and Asynchronous FIFO Design with DFT Insertion | Synopsys VCS, DC Compiler**
Developed and synthesized adaptable single and dual-clock FIFO designs with SAED90nm DFT cells, incorporating configurable width and depth features. Validated functionality and tested various scenarios using simulation waveforms in Synopsys VCS.
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CERTIFICATIONS

- Completed a course “**System Design through Verilog**” from NPTEL (National Programme on Technology Enhanced Learning), IIT (Indian Institute of Technology).